Assignment

Computer_Organization_architecture

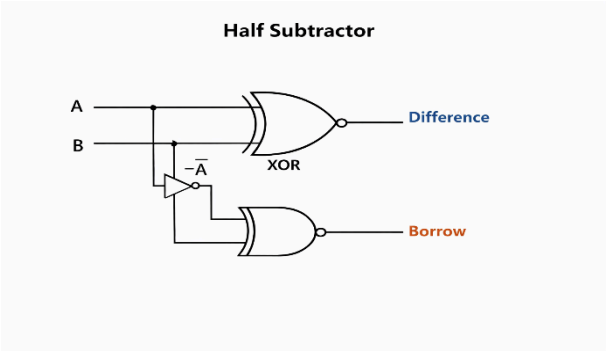
06-04-2026

Q.1 Write Verilog code and its test bench for half subtractor?

Half Subtractor Logic

A half subtractor subtracts two 1-bit inputs:

- Inputs: **A (minuend), B (subtrahend)**
- Outputs:
  - **Difference = A ⊕ B**
  - **Borrow = $\bar{A}$ · B** (NOT A AND B)



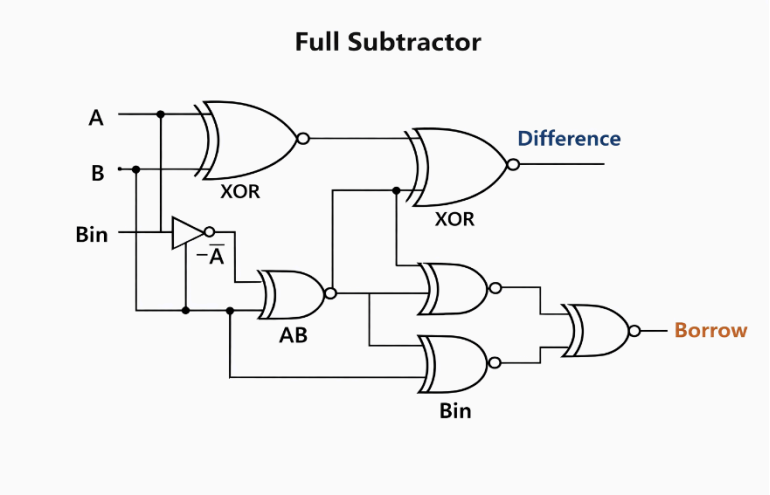
Q.2 Write Verilog code and its test bench for full subtractor

A **Full Subtractor** subtracts three inputs:

- **A** → Minuend
- **B** → Subtrahend
- **Bin** → Borrow input

Outputs:

- **Difference = A ⊕ B ⊕ Bin**
- **Borrow = (¬A · B) + (¬A · Bin) + (B · Bin)**



Q3. Design and Verify a D Flip-Flop

Design a positive edge-triggered D Flip-Flop with asynchronous active-high reset.

Requirements:

1. Write the Verilog/SystemVerilog code for the D Flip-Flop.
2. Write a testbench to verify the following cases:
    - Reset behavior
    - Data capture on clock edge
    - Hold behavior between clock edges
3. Simulate and attach the waveform.

Given the following inputs:

Time (ns)	clk	reset	d
0	0	1	X
5	↑	0	1
15	↑	0	0
25	↑	0	1